US012400698B2

(12) United States Patent
van der Straten et al.

(10) Patent No.: US 12,400,698 B2
(45) Date of Patent: Aug. 26, 2025

(54) **MRAM DEVICE WITH OCTAGON PROFILE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 183 days.

(21) Appl. No.: **18/048,455**

(22) Filed: **Oct. 21, 2022**

(65) **Prior Publication Data**

US 2024/0135978 A1 Apr. 25, 2024
US 2024/0233793 A9 Jul. 11, 2024

(51) **Int. Cl.**
H10N 50/80 (2023.01)
G11C 11/16 (2006.01)

(52) **U.S. Cl.**
CPC ***G11C 11/161*** (2013.01)

(58) **Field of Classification Search**
CPC G11C 11/161; H10B 61/00; H10N 50/80; H10N 50/10
See application file for complete search history.

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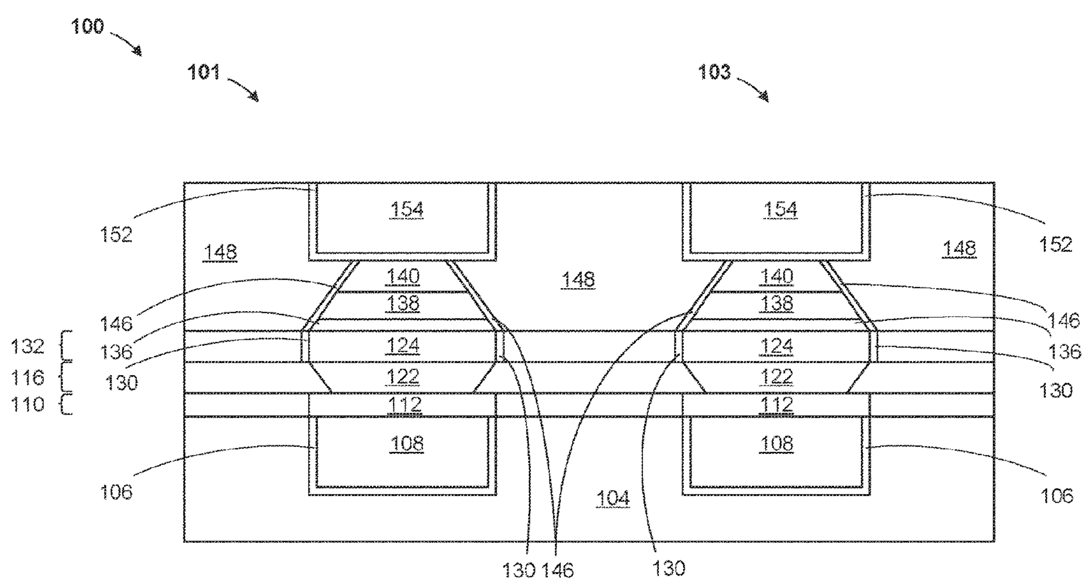
(57) **ABSTRACT**

A magnetic tunnel junction cell, a cross section with octagon profile, vertically aligned layers of a top electrode, a free layer, a tunneling barrier, a reference layer, a bottom electrode with tapered side surface with a width at an upper surface greater than a width at a lower surface, the reference layer with vertical side surface perpendicular to an upper horizontal surface of the bottom electrode, the free layer and the tunneling barrier, each include a tapered side surface of the same angle, and each include a width at an upper surface less than a width at a lower surface. Forming a bottom electrode of a magnetic tunnel junction cell with a tapered side surface with a width at an upper surface greater than a width at a lower surface, forming a reference layer with vertical side surface perpendicular to an upper surface of the bottom electrode.

13 Claims, 14 Drawing Sheets



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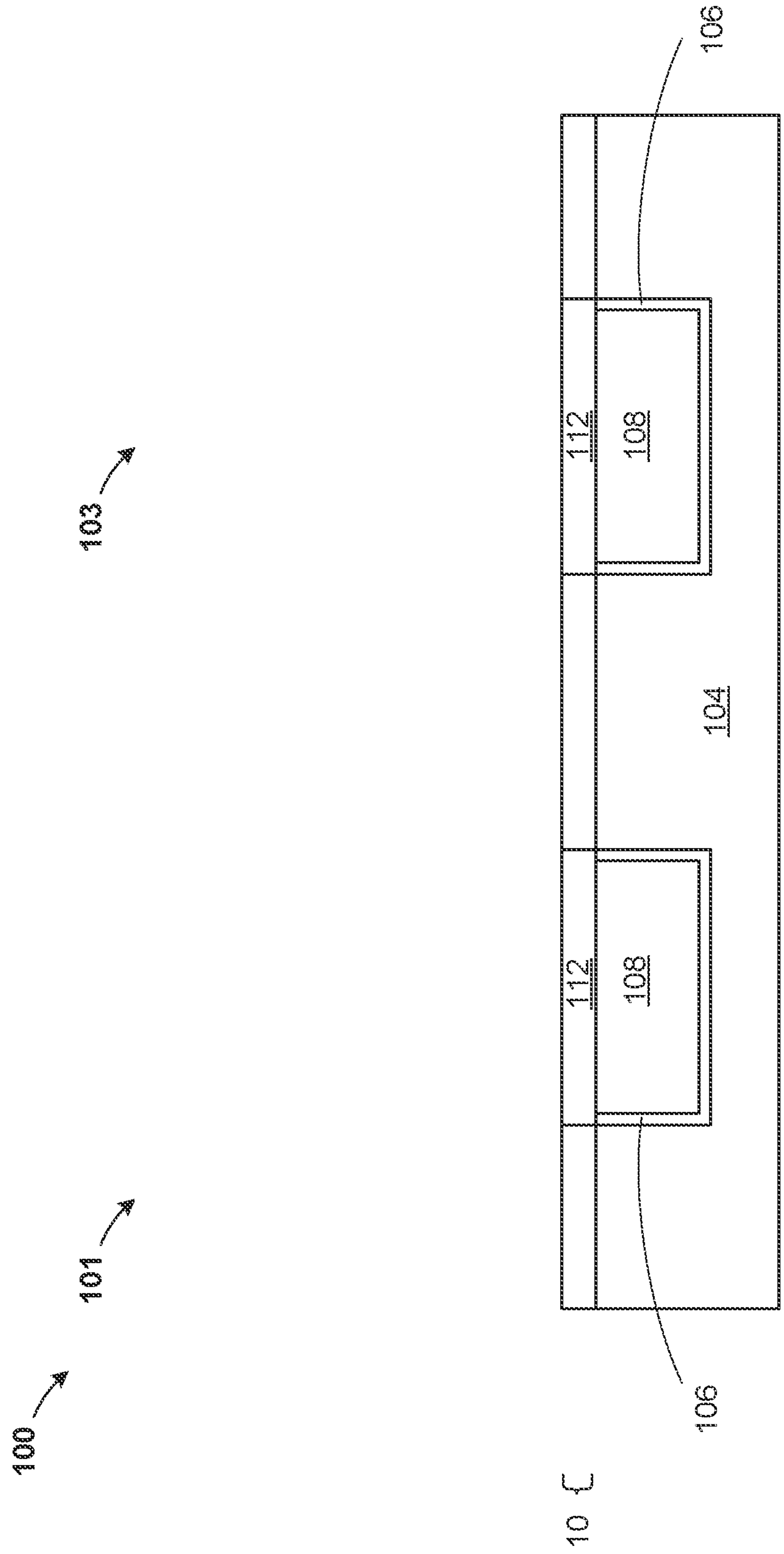
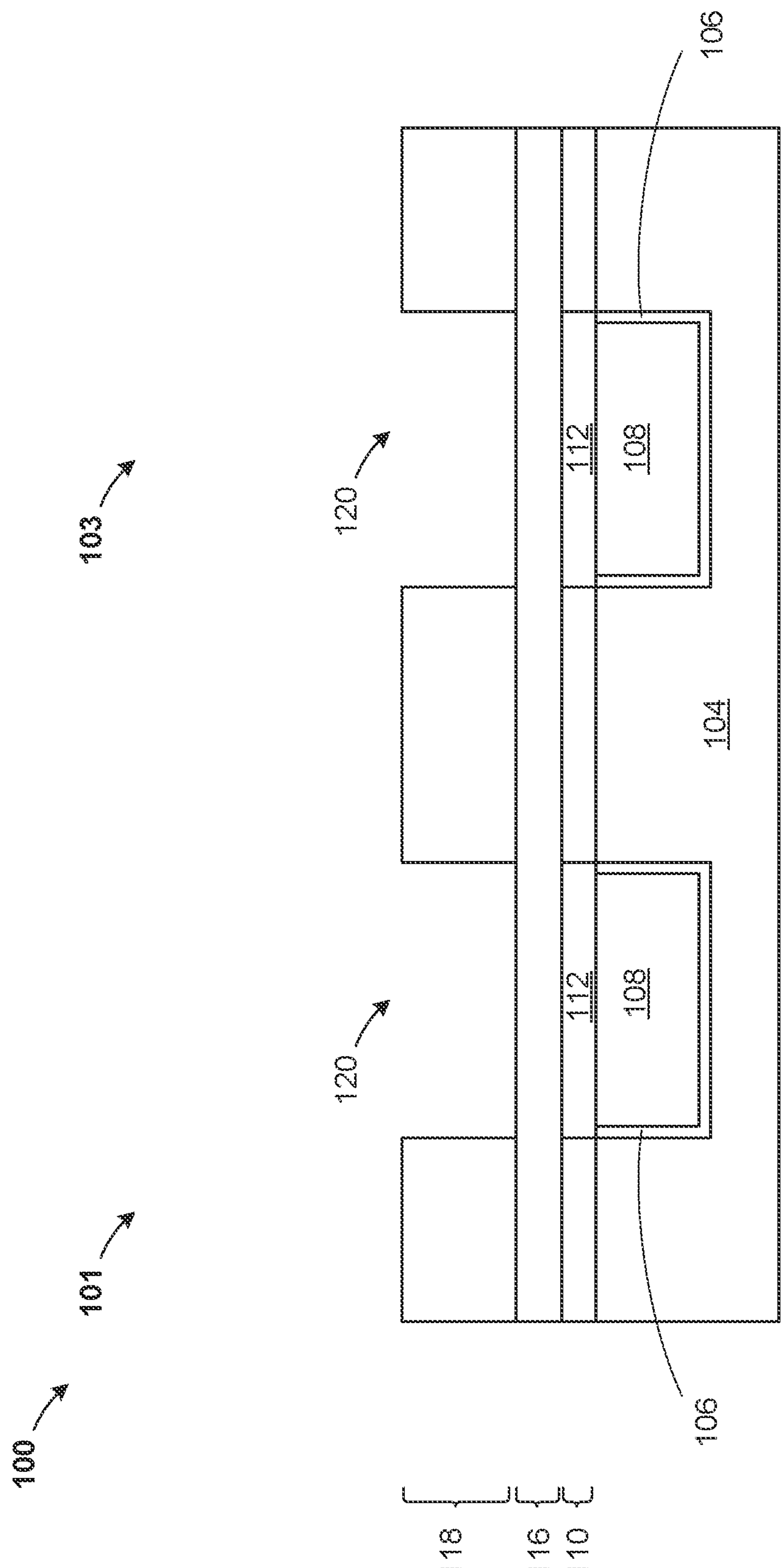


Figure 1



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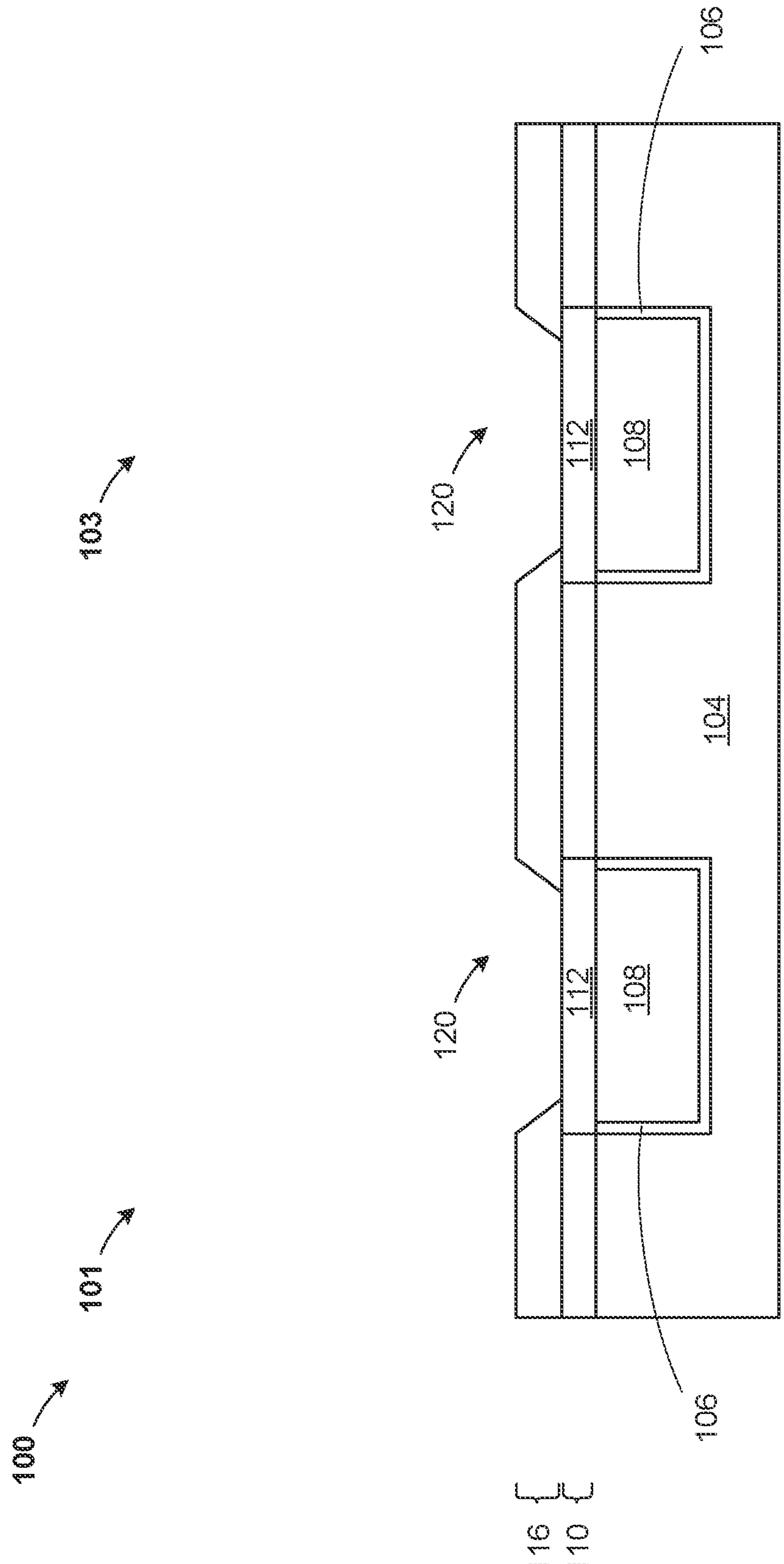

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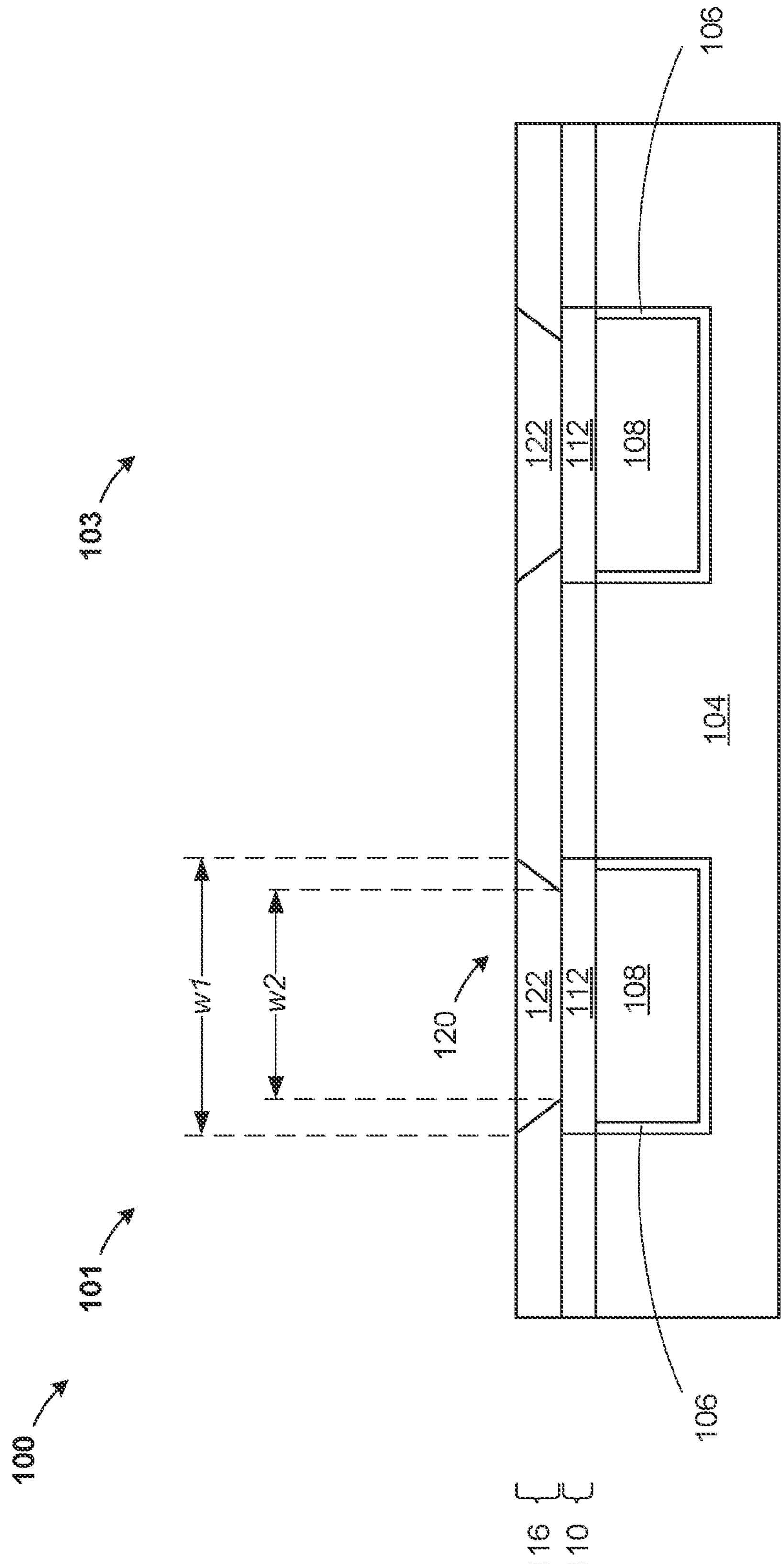

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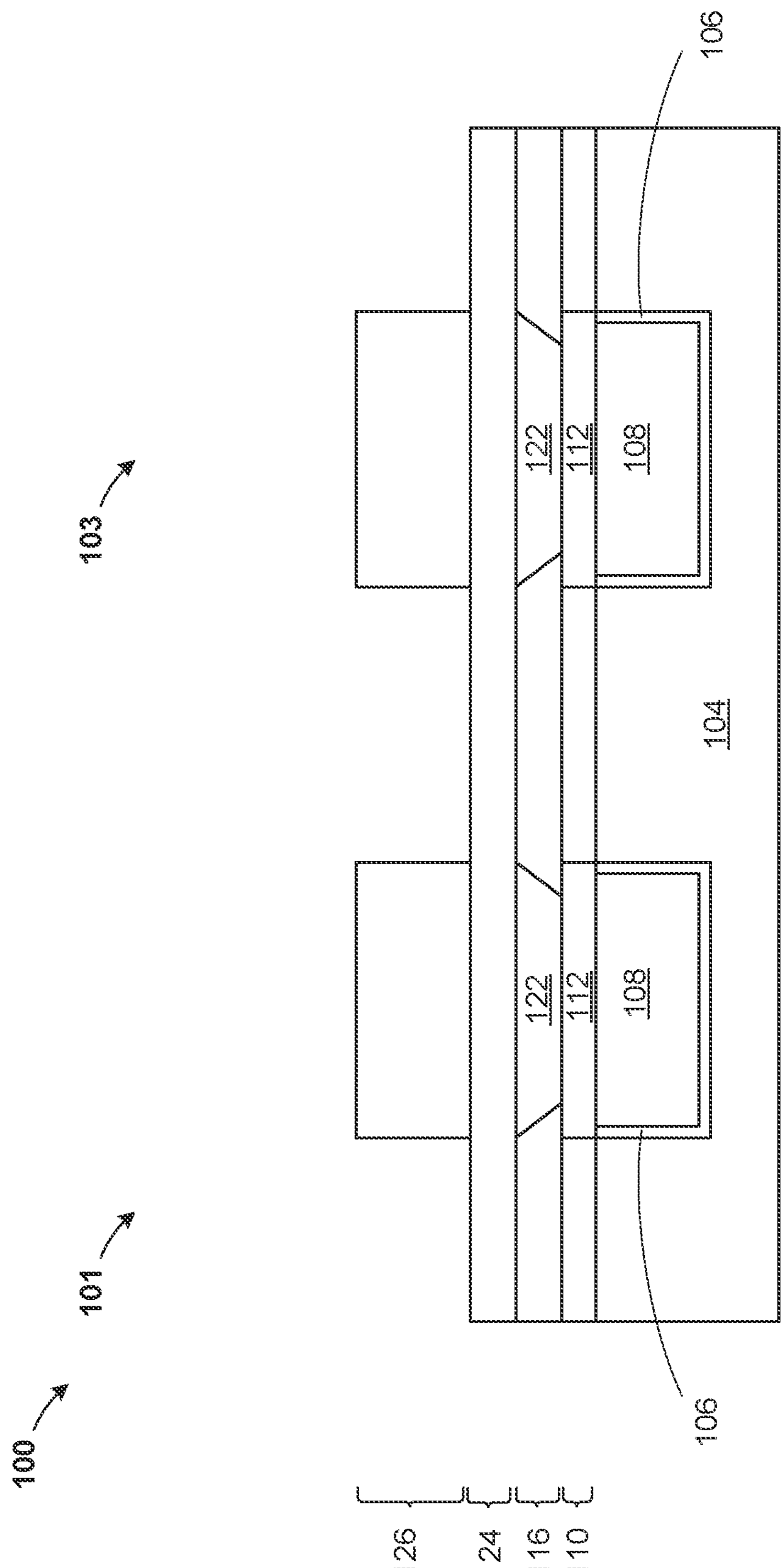

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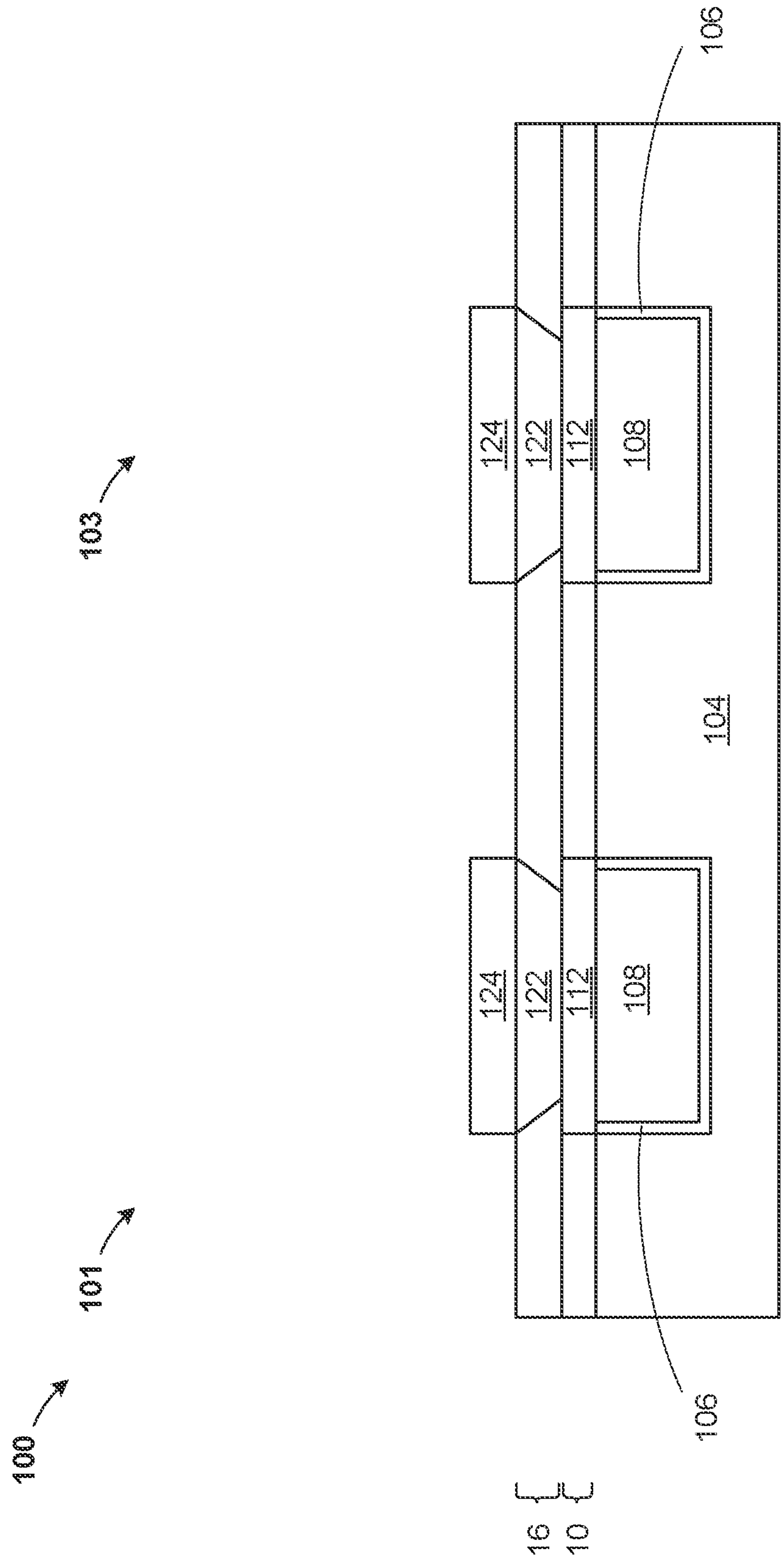

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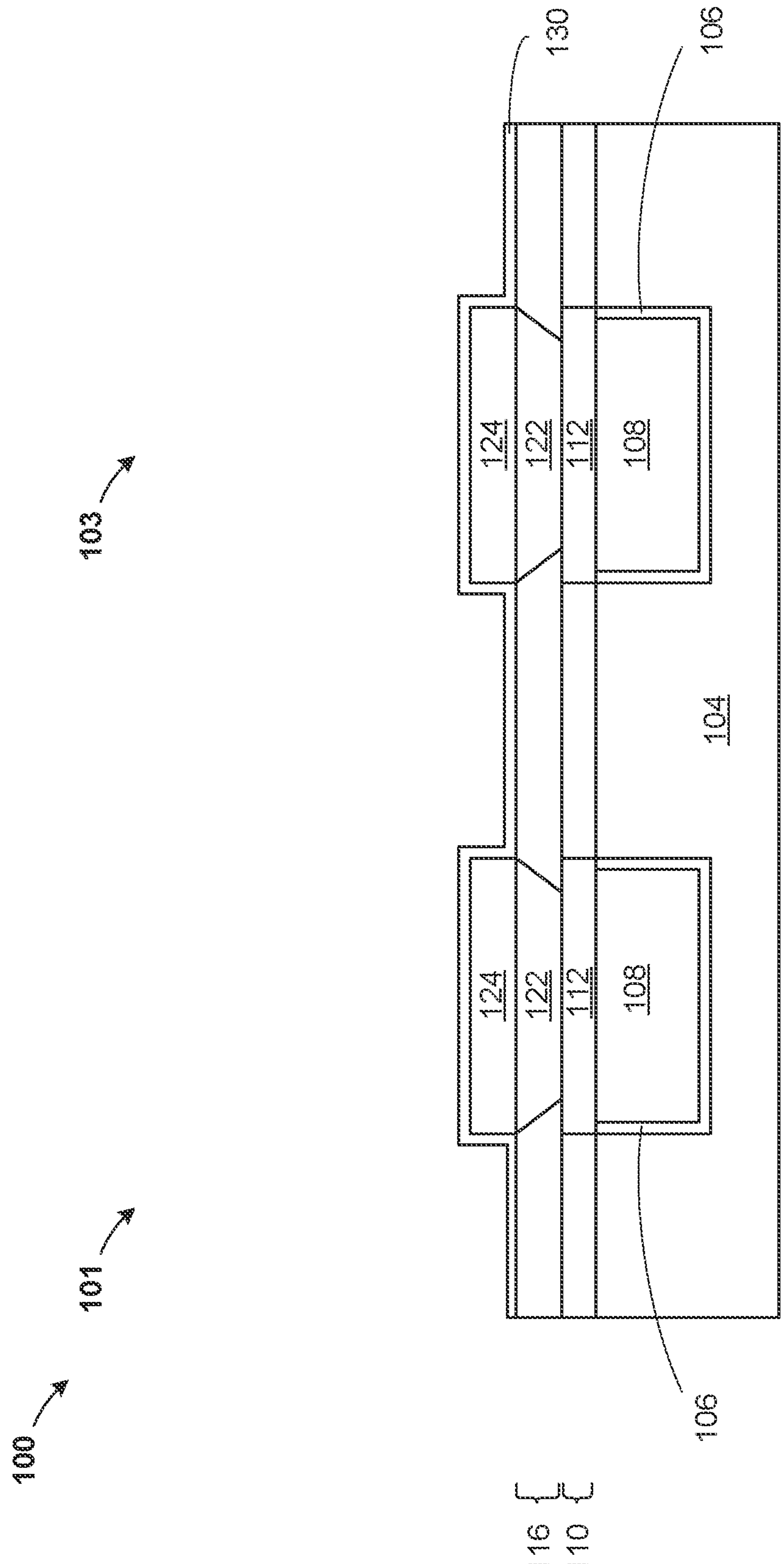

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Figure 7

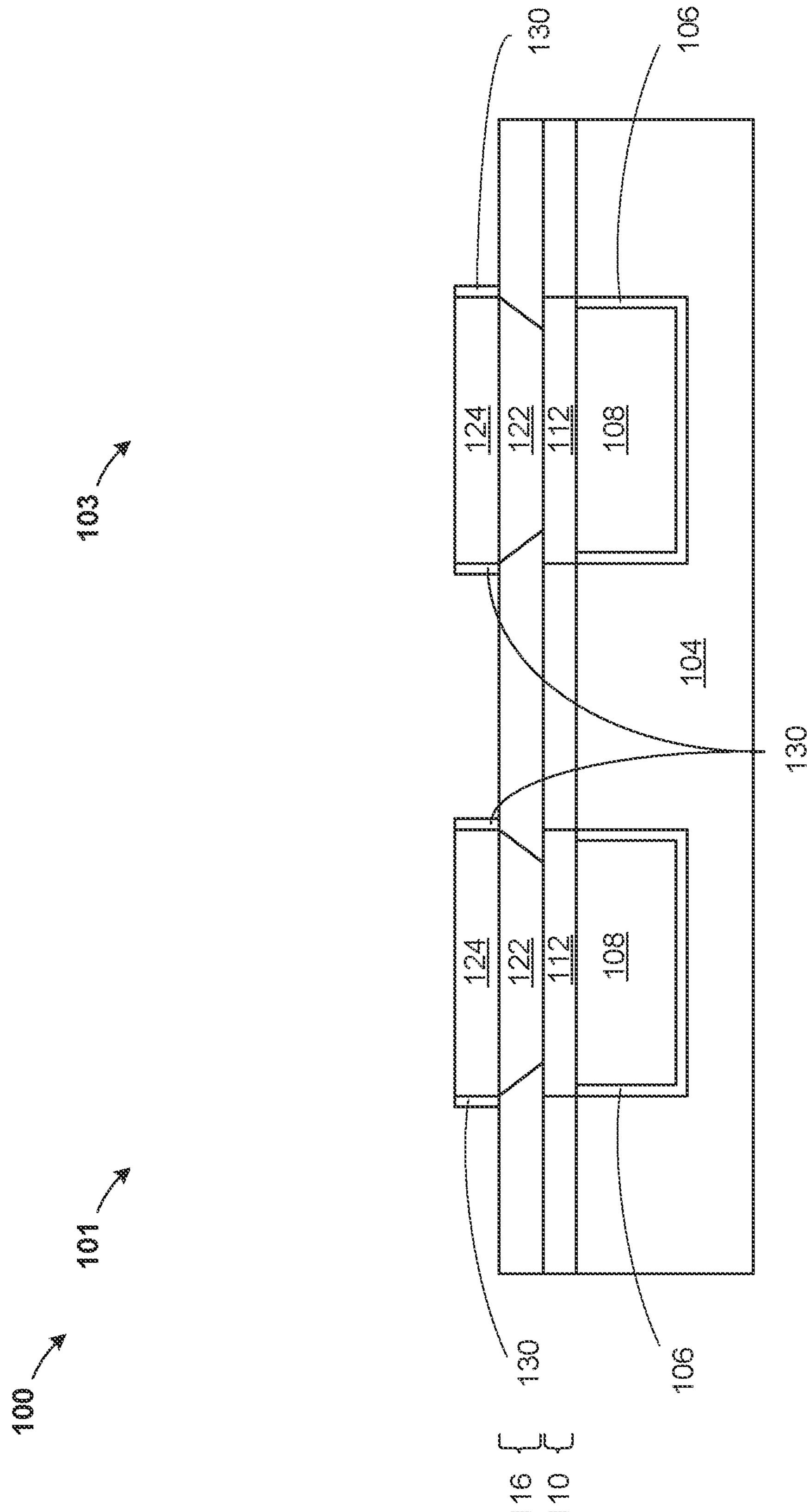

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Figure 8

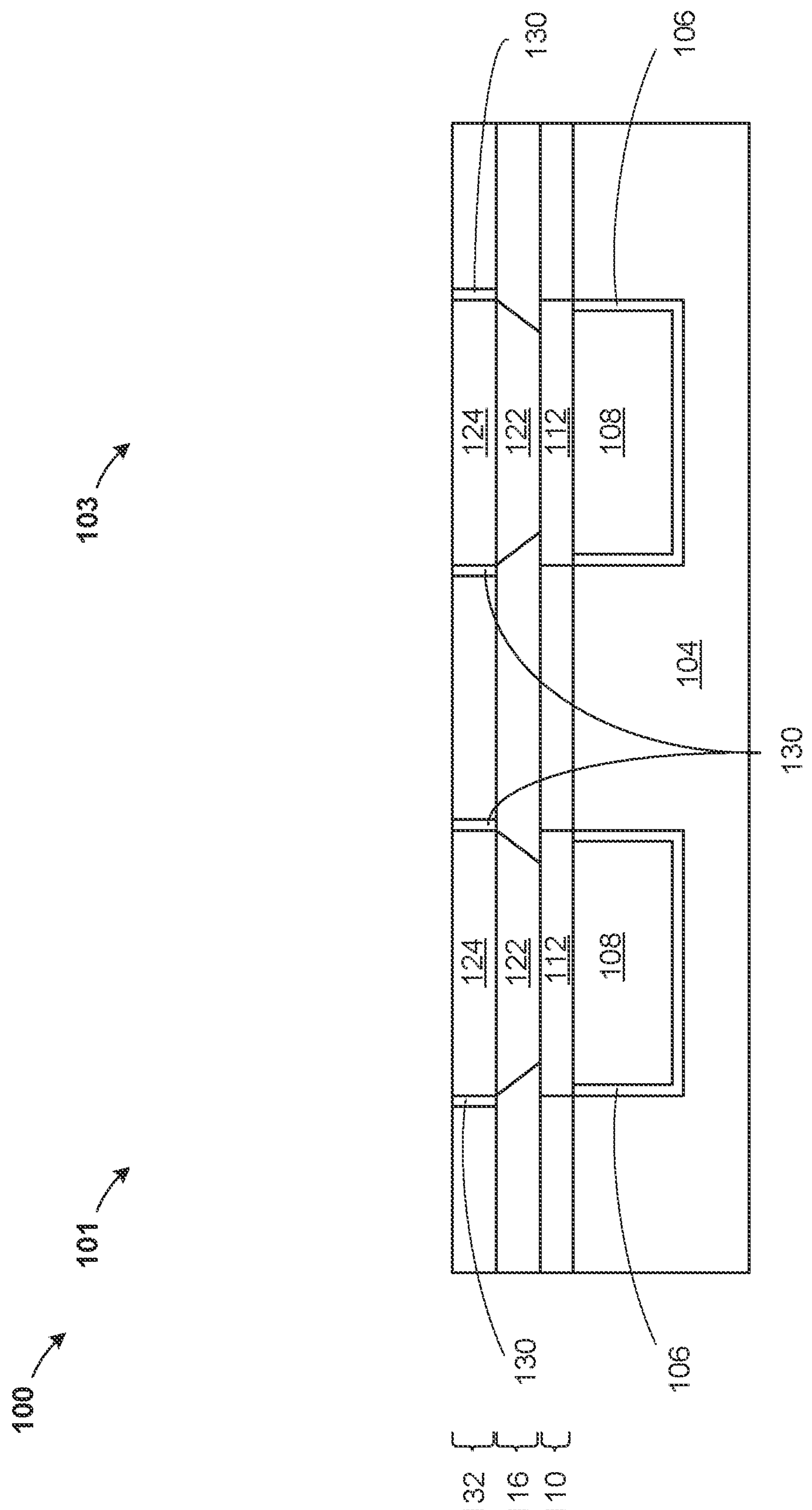
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Figure 9

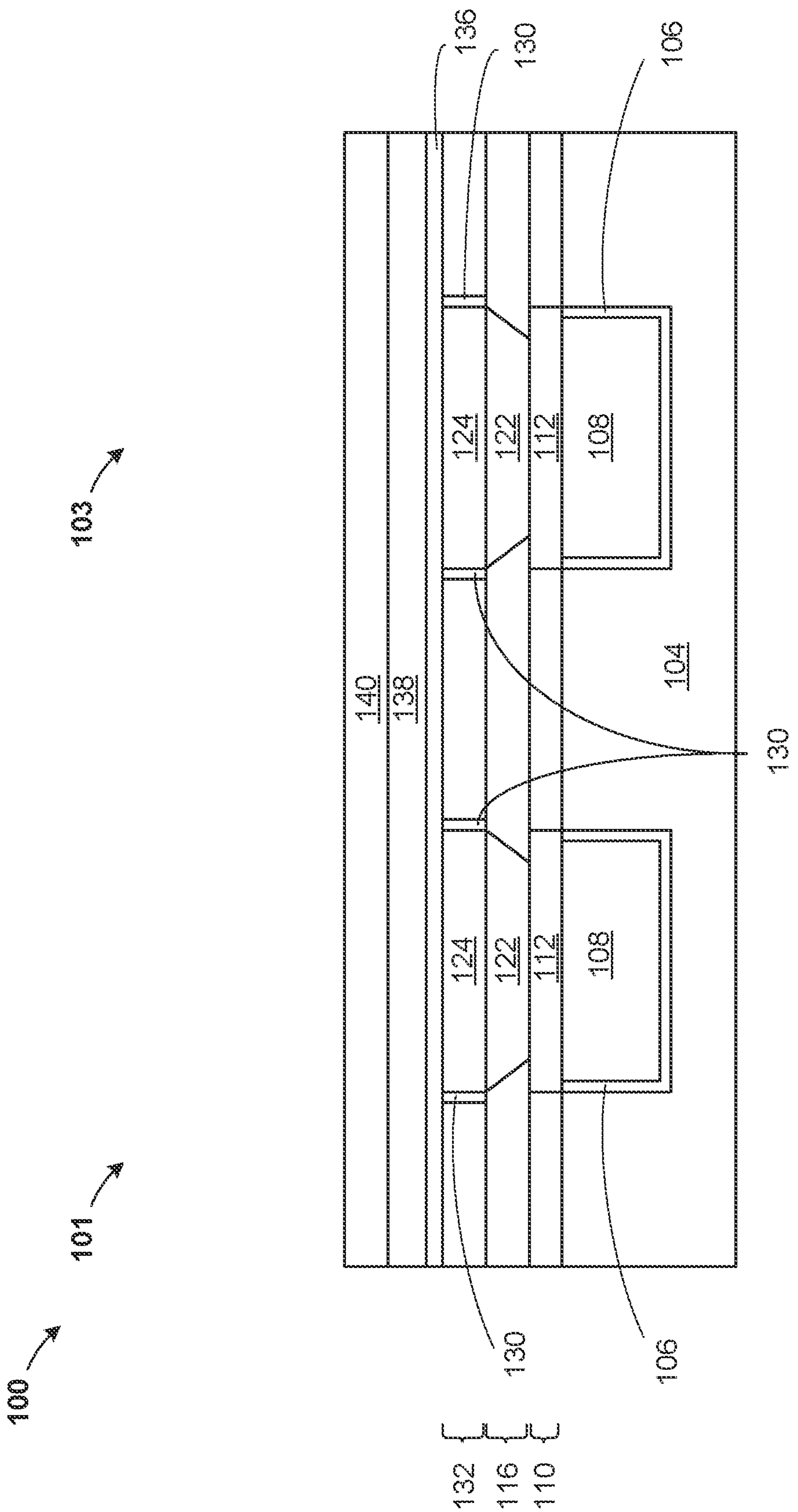
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Figure 10

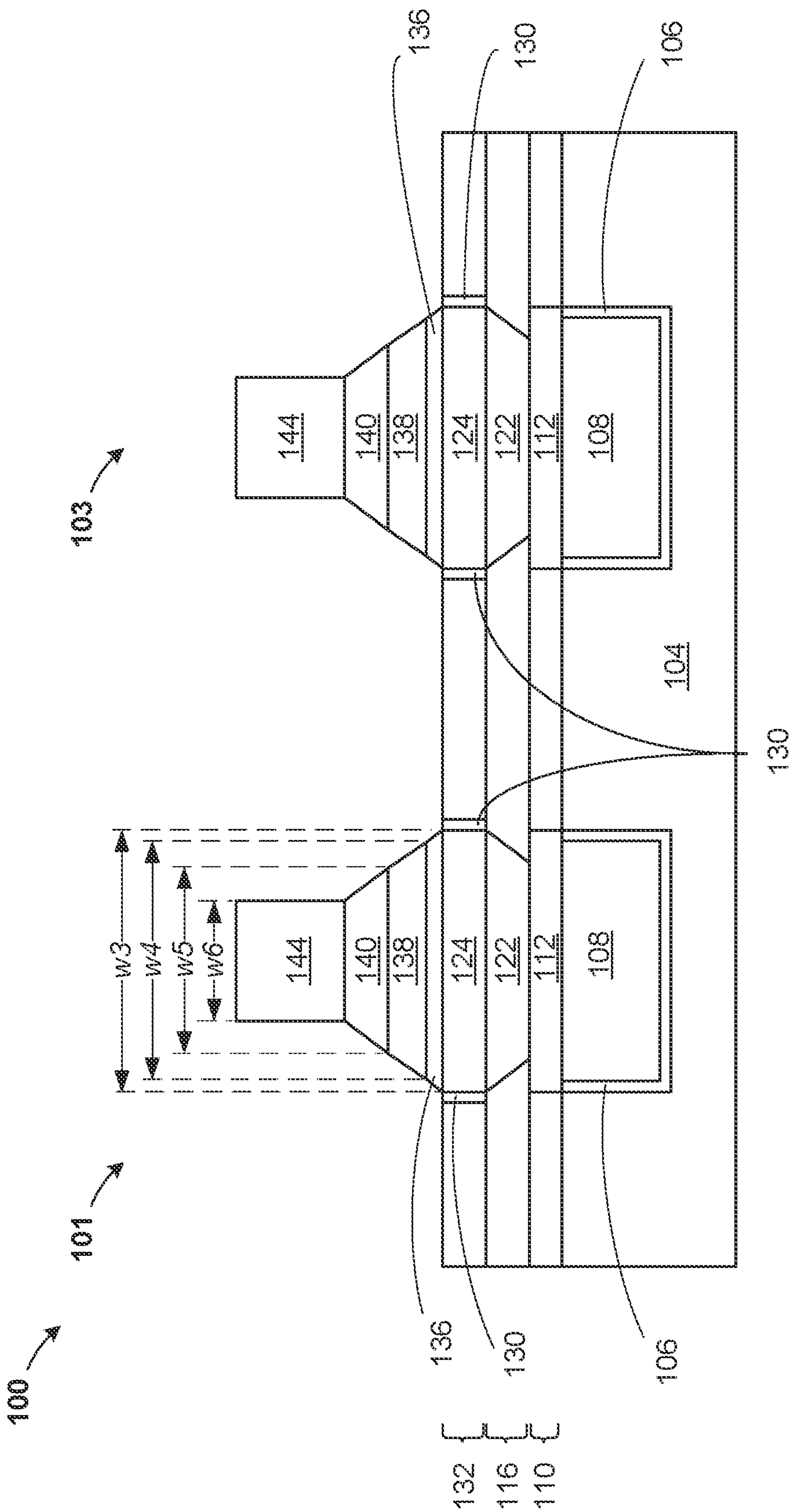
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Figure 11

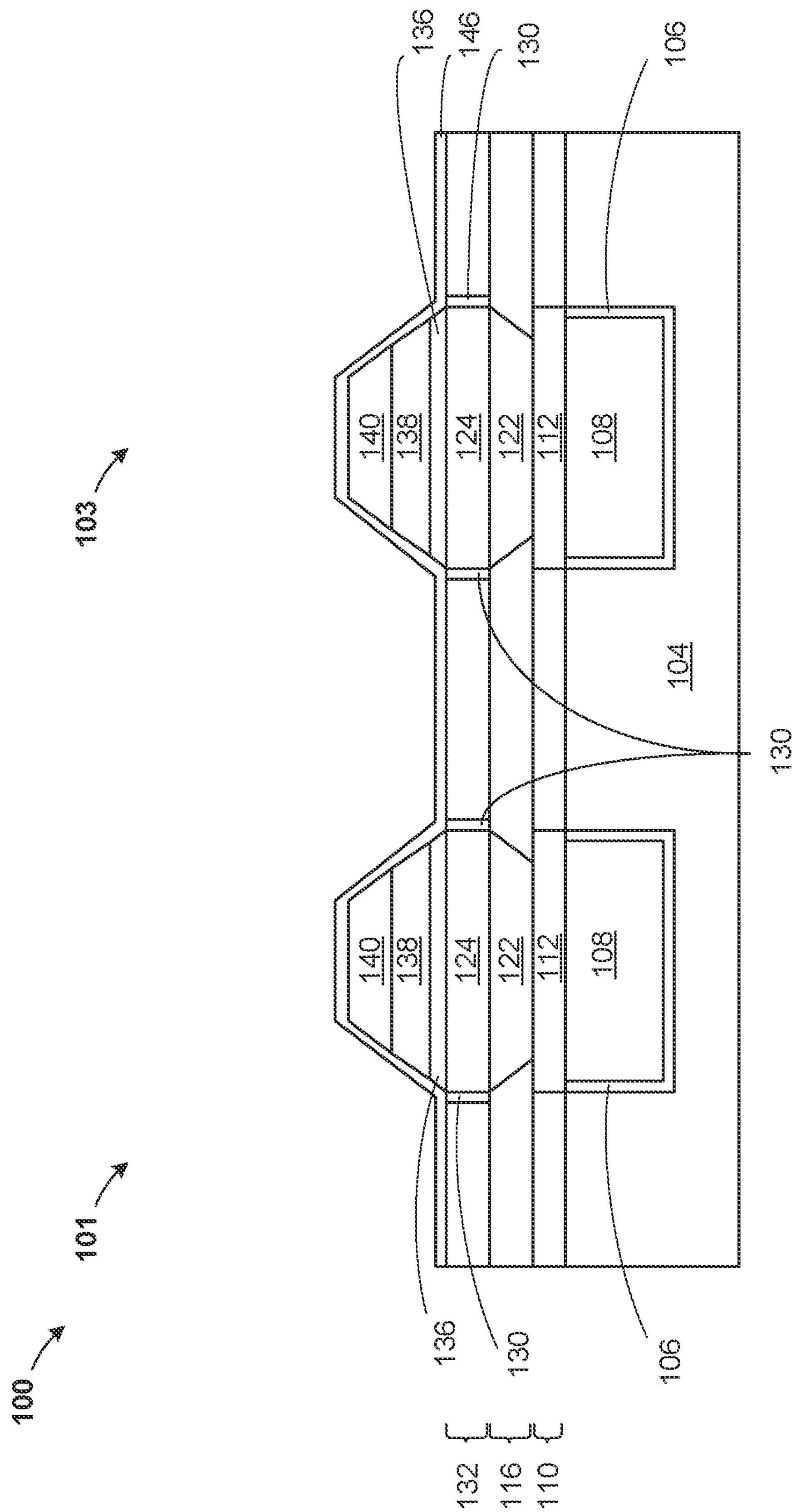
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Figure 12

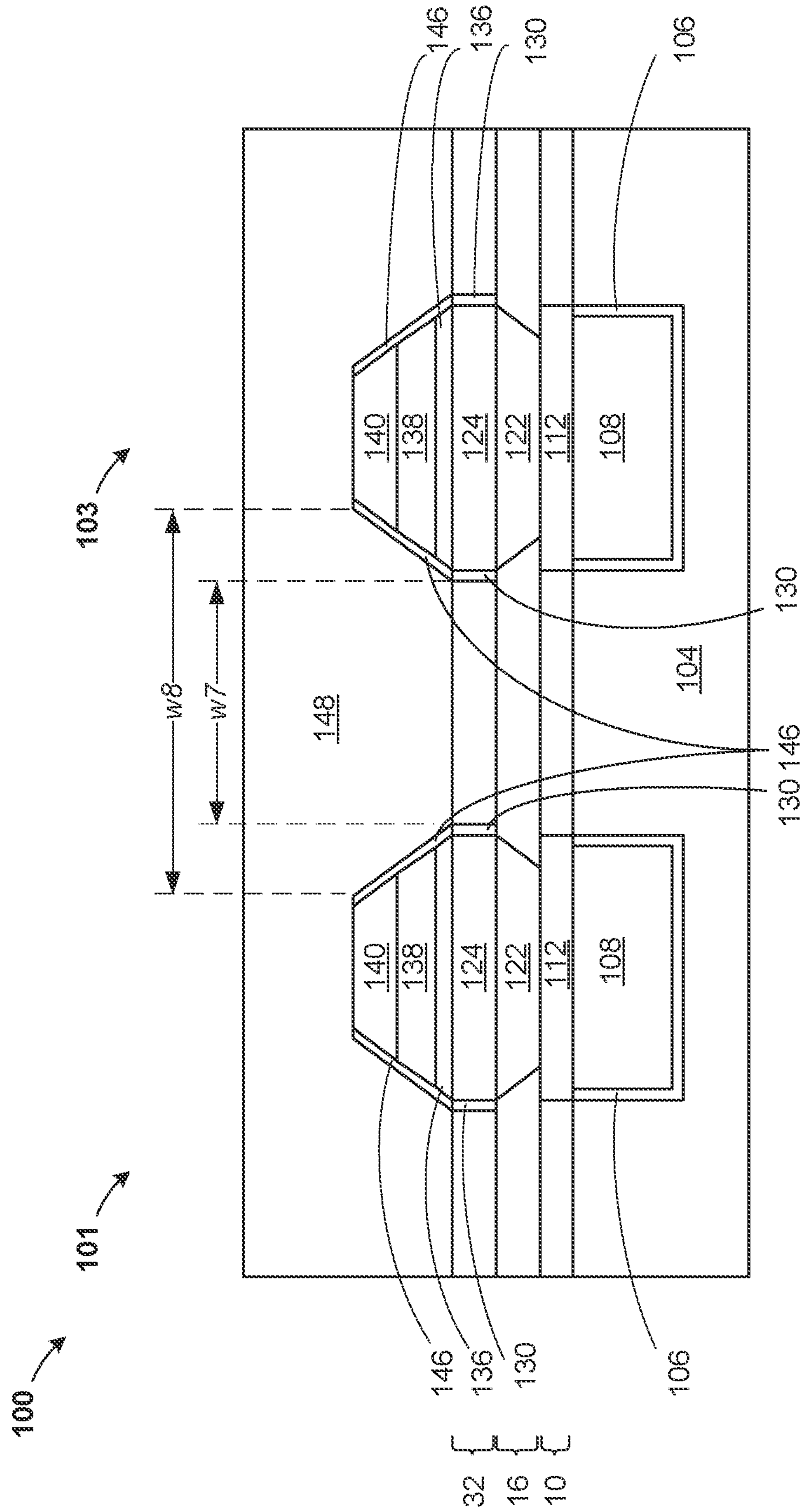
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Figure 13

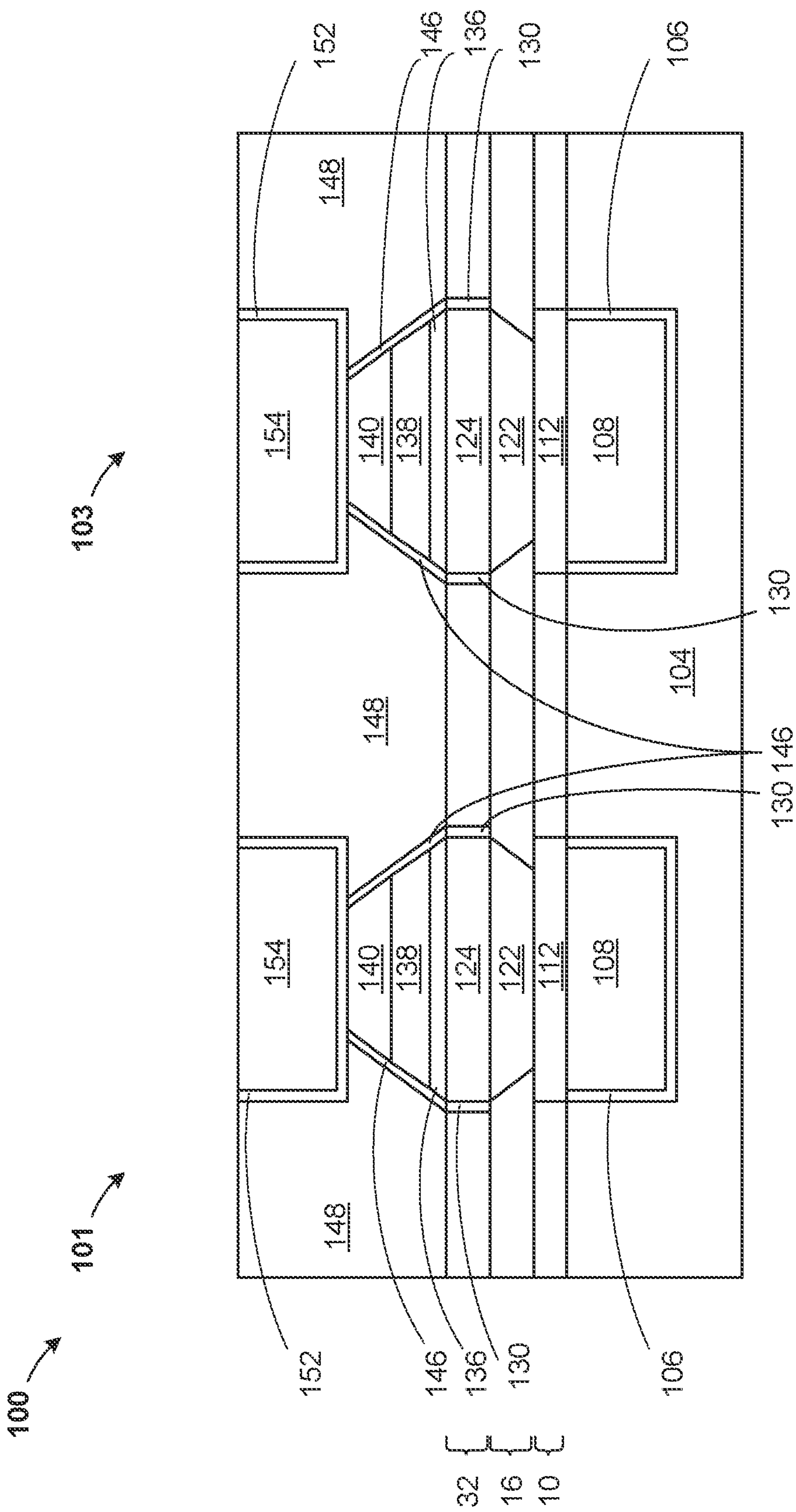
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Figure 14

MRAM DEVICE WITH OCTAGON PROFILE

BACKGROUND

The present invention relates, generally, to the field of semiconductor manufacturing, and more particularly to fabricating a magnetic tunnel junction device with an octagon profile.

Magneto resistive random-access memory ("MRAM") devices are used as non-volatile computer memory. MRAM data is stored by magnetic storage elements. The elements are formed from two ferromagnetic layers, each of which can hold a magnetic field, separated by a spin conductor layer. One of the two layers is a reference magnet, or a reference layer, set to a particular polarity, while the remaining layer's field can be changed to match that of an external field to store memory and is termed the "free magnet" or "free-layer". This configuration is known as the magnetic tunnel junction (MTJ) and is the simplest structure for a MRAM bit of memory.

SUMMARY

According to an embodiment of the present invention, a semiconductor device is provided. The semiconductor device including a magnetic tunnel junction (MTJ) stack, where a cross section of the MTJ stack includes an octagon profile.

According to an embodiment of the present invention, a semiconductor device is provided. The semiconductor device including a magnetic tunnel junction (MTJ) stack, where a vertical side surface of the MTJ stack comprises an octagon profile, the MTJ stack includes vertically aligned layers of a top electrode, a free layer, a tunneling barrier, a reference layer and a bottom electrode, the bottom electrode includes a tapered side surface including a width at an upper surface of the bottom electrode greater than a width at a lower surface of the bottom electrode, the reference layer includes a vertical side surface perpendicular to an upper horizontal surface of the bottom electrode and the top electrode, the free layer and the tunneling barrier, each include a tapered side surface of the same angle, and each include a width at an upper surface less than a width at a lower surface.

According to an embodiment of the present invention, a method is provided. The method including forming a bottom electrode of a magnetic tunnel junction (MTJ) stack in a first dielectric, the bottom electrode includes a tapered side surface including a width at an upper surface of the bottom electrode greater than a width at a lower surface of the bottom electrode and forming a reference layer of the first magnetic tunnel function (MTJ) stack, the reference layer includes a vertical side surface perpendicular to an upper horizontal surface of the bottom electrode.

BRIEF DESCRIPTION OF THE DRAWINGS

The following detailed description, given by way of example and not intended to limit the invention solely thereto, will best be appreciated in conjunction with the accompanying drawings, in which:

FIG. **1** illustrates a cross-sectional view of a multi-state memory cell, according to an exemplary embodiment;

FIG. **2** illustrates a cross-sectional view of the multi-state memory cell and illustrates formation of an inter-layer dielectric and a hard mask, according to an exemplary embodiment;

FIG. **3** illustrates a cross-sectional view of the multi-state memory cell and illustrates patterning of the inter-layer dielectric and removal of the hard mask, according to an exemplary embodiment;

FIG. **4** illustrates a cross-sectional view of the multi-state memory cell and illustrates formation of a bottom electrode, according to an exemplary embodiment;

FIG. **5** illustrates a cross-sectional view of the multi-state memory cell and illustrates formation of a reference layer and a hard mask, according to an exemplary embodiment;

FIG. **6** illustrates a cross-sectional view of the multi-state memory cell and illustrates patterning of the reference layer and removal of the hard mask, according to an exemplary embodiment;

FIG. **7** illustrates a cross-sectional view of the multi-state memory cell and illustrates formation of a first encapsulation layer, according to an exemplary embodiment;

FIG. **8** illustrates a cross-sectional view of the multi-state memory cell and illustrates removal of portions of the first encapsulation layer, according to an exemplary embodiment;

FIG. **9** illustrates a cross-sectional view of the multi-state memory cell and illustrates formation of an inter-layer dielectric, according to an exemplary embodiment;

FIG. **10** illustrates a cross-sectional view of the multi-state memory cell and illustrates formation of a tunneling barrier and a free layer and a top electrode, according to an exemplary embodiment;

FIG. **11** illustrates a cross-sectional view of the multi-state memory cell and illustrates formation of a hard mask and patterning of the tunneling barrier, the free layer and the top electrode, according to an exemplary embodiment;

FIG. **12** illustrates a cross-sectional view of the multi-state memory cell and illustrates removal of the hard mask and formation of a second encapsulation layer, according to an exemplary embodiment;

FIG. **13** illustrates a cross-sectional view of the multi-state memory cell and illustrates removal of portions of the first encapsulation layer and formation of an inter-layer dielectric, according to an exemplary embodiment; and

FIG. **14** illustrates a cross-sectional view of the multi-state memory cell and illustrates formation of an upper metal wire, according to an exemplary embodiment.

The drawings are not necessarily to scale. The drawings are merely schematic representations, not intended to portray specific parameters of the invention. The drawings are intended to depict only typical embodiments of the invention. In the drawings, like numbering represents like elements.

DETAILED DESCRIPTION

Detailed embodiments of the claimed structures and methods are disclosed herein; however, it can be understood that the disclosed embodiments are merely illustrative of the claimed structures and methods that may be embodied in various forms. This invention may, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiment set forth herein. Rather, these exemplary embodiments are provided so that this disclosure will be thorough and complete and will fully convey the scope of this invention to those skilled in the art. In the description, details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the presented embodiments.

For purposes of the description hereinafter, the terms "upper", "lower", "right", "left", "vertical", "horizontal", "top", "bottom", and derivatives thereof shall relate to the disclosed structures and methods, as oriented in the drawing figures. The terms "overlying", "atop", "on top", "positioned on" or "positioned atop" mean that a first element, such as a first structure, is present on a second element, such as a second structure, wherein intervening elements, such as an interface structure may be present between the first element and the second element. The term "direct contact" means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary conducting, insulating or semiconductor layers at the interface of the two elements.

In the interest of not obscuring the presentation of embodiments of the present invention, in the following detailed description, some processing steps or operations that are known in the art may have been combined together for presentation and for illustration purposes and in some instances may have not been described in detail. In other instances, some processing steps or operations that are known in the art may not be described at all. It should be understood that the following description is rather focused on the distinctive features or elements of various embodiments of the present invention.

As stated above, magneto resistive random-access memory (hereinafter "MRAM") devices are a non-volatile computer memory technology. MRAM data is stored by magnetic storage elements. The elements are formed from two ferromagnetic layers, each of which can hold a magnetic field, separated by a spin conductor layer. One of the two layers is a reference magnet, or a reference layer, set to a particular polarity, while the remaining layer's field can be changed to match that of an external field to store memory and is termed the "free magnet" or "free-layer". The magnetic reference layer may be referred to as a reference layer, and the remaining layer may be referred to as a free layer. This configuration is known as the magnetic tunnel junction (hereinafter "MTJ") and is the simplest structure for a MRAM bit of memory.

A memory device is built from a grid of such memory cells or bits. In some configurations of MRAM, such as the type further discussed herein, the magnetization of the magnetic reference layer is fixed in one direction (up or down), and the direction of the magnetic free layer can be switched by external forces, such as an external magnetic field or a spin-transfer torque generating charge current. A smaller current (of either polarity) can be used to read resistance of the device, which depends on relative orientations of the magnetizations of the magnetic free layer and the magnetic reference layer. The resistance is typically higher which the magnetizations are anti-parallel and lower when they are parallel, though this can be reversed, depending on materials used in fabrication of the MRAM.

The MRAM stack layers may be conformally formed using known techniques. In formation of the MTJ stacks layers, the reference layer is formed on a dielectric and a bottom electrode. The tunneling barrier layer is formed on the reference layer. In an embodiment, the tunneling barrier layer is a barrier, such as a thin insulating layer or electric potential, between two electrically conducting materials. Electrons (or quasiparticles) pass through the tunneling barrier layer by the process of quantum tunneling. In certain embodiments, the tunneling barrier layer includes at least one sublayer composed of magnesium oxide (MgO). It should be appreciated that materials other than MgO can be used to form the tunneling barrier layer. The free layer is a magnetic free layer that is adjacent to tunneling barrier layer and opposite the reference layer. The free layer has a magnetic moment or magnetization that can be flipped. It should also be appreciated that the MTJ stack layers may include additional layers, omit certain layers, and each of the layers may include any number of sublayers. Moreover, the composition of layers and/or sublayers may be different between the different MRAM stacks.

For high performance MRAM devices based on perpendicular magnetic tunnel junction (MTJ) structures, well-defined interfaces and interface control are essential. MTJ structures typically include a cobalt (Co) based synthetic anti-ferromagnet (SAF), a CoFeB-based reference layer, a MgO-based tunnel barrier, a CoFeB-based free layer, and cap layers containing e.g. tantalum (Ta) and/or ruthenium (Ru). Embedded MTJ structures are usually formed by subtractive patterning of blanket MTJ stacks into pillars between two metal levels. In this invention, after each layer of MTJ stack patterning, inter-pillar spaces are filled with inter-layer dielectric (hereinafter "ILD") to enable connection to BEOL wiring by a top contact level without voids in the ILD. ILD gap fill between pillars presents a significant challenge since the presence of voids in the ILD between the pillars can lead to shorts.

The present invention relates to fabricating a MTJ device with outwardly tapered vertical side surface of the top electrode, the free layer and the tunneling barrier, when moving from an upper surface to a lower surface of the MTJ device, and an inwardly tapered vertically side surface of a bottom electrode, when moving from an upper surface to a lower surface of the MTJ device, resulting in an octagon side profile of the MTJ device. Forming the octagon side profile of the MTJ device helps to reduce gaps in the ILD surrounding the devices as three separately formed ILDs, each for a lower height than forming a single ILD, surround an entire vertical side surface of the MTJ device. Additionally, when forming the uppermost ILD, a lower surface of the uppermost ILD has a smaller area than an upper surface of the uppermost ILD, helping to reduce formation of voids. The MTJ device with individually formed layers improved MRAM performance due to reduced shorts between top contacts of adjacent MTJ device pillars.

For high performance MRAM devices based on perpendicular magnetic tunnel junction (MTJ) structures, well-defined interfaces and interface control are essential. MTJ structures typically include a Co-based synthetic anti-ferromagnet (SAF), a CoFeB-based reference layer, a MgO-based tunnel barrier, a CoFeB-based free layer, and cap layers containing e.g. Ta and/or Ru. Embedded MTJ structures are usually formed by subtractive patterning of blanket MTJ stacks into pillars between two metal levels. After MTJ stack patterning, the inter-pillar spaces are filled with an inter-layer dielectric (ILD) to enable connection to BEOL wiring by a top contact level. ILD gap fill between pillars presents a significant challenge since the presence of voids in the ILD between the pillars can lead to shorts. When forming the ILD, voids may occur between adjacent MTJ stacks. The voids may then become filled with material intended to form a top contact for each MTJ stack. The filled voids may cause shorts between adjacent top contacts of adjacent MTJ stacks. There is a need for embedded MTJ structures formed with a reduced risk of ILD void induced shorts.

An MRAM pillar having an octagon profile helps to reduce ILD voiding between adjacent MTJ stacks. This helps to extend scalability of MRAM device memory elements due to void-free gap fill between MRAM pillars and improved embedded MRAM performance due to reduced top contact shorts.

In an embodiment, a bottom electrode may be formed on a lower metal wire in an opening of a first inter-layer dielectric. The bottom electrode may have a wider upper surface than a lower surface, such that sidewalls of the bottom electrode are angled or flare outward. A reference layer may be formed conformally and then patterned such that sidewalls of the reference layer are generally vertically aligned above the bottom electrode, and perpendicular to an upper surface of the lower metal wire. The vertical sidewall of the reference layer may have a sidewall angle, with a value of 90 degrees in an embodiment. A first encapsulation layer may be formed and patterned such that the first encapsulation layer surrounds vertical sidewalls of the reference layer. A second inter-layer dielectric may be formed surrounding the reference layer and the first encapsulation layer of each MTJ stack. A tunneling barrier, a free layer and material for a top electrode may each be conformally formed on the second inter-layer dielectric, the first encapsulation layer and the reference layer. Portions of the tunneling barrier, the free layer and the material for the top electrode may be removed such that sidewalls of the tunneling barrier, the free layer and the top electrode remain in each MTJ stack. Each layer of the tunneling barrier, the free layer and the top electrode have a wider upper surface than a lower surface, or angled or flared outward when traveling from an upper most surface to a lower surface of the MTJ stack. Vertical sidewalls of the tunneling barrier, the free layer and the top electrode may each have the same sidewall angle, which may range between 10 and 50 degrees in an embodiment. A second encapsulation layer may be formed and patterned such that the first encapsulation layer surrounds vertical sidewalls of the tunneling barrier, the free layer and the top electrode. A third inter-layer dielectric may be formed surrounding the tunneling barrier and the free layer and above the MTJ stacks. An upper metal wire may be formed in vertically aligned openings in the inter-layer dielectric above the top electrode, the free layer, the tunneling barrier the reference layer, the bottom electrode and the lower metal wire. The resulting MTJ stack has an octagon side profile. Vertical side surfaces of the top electrode, the free layer and the tunneling barrier may flare outward, when moving from an upper surface to a lower surface of the MTJ stack. The reference layer may have a vertical side surface which is essentially perpendicular to a lower surface of the MTJ stack. The bottom electrode may have a vertical side surface which flares inward, when moving from an upper surface to a lower surface of the MTJ stack.

In an embodiment, the first and the second encapsulation layer may contain the same material. In an alternate embodiment, the first and the second encapsulation layer may each contain different materials. In an embodiment, either or both of the first and the second encapsulation layer may contain silicon nitride (SiN) or silicon nitride carbon (SiNC). In an embodiment, the first encapsulation layer may contain zirconium oxide (ZrO_2). Materials for the first, second and third inter-layer dielectric may contain the same material. In an alternate embodiment, the first, second and third inter-layer dielectric may each contain different material. The first, second and third inter-layer dielectric may each include SiCOH, SiCNO and SiCHNO.

Referring now to FIG. **1**, a semiconductor structure **100** (hereinafter "structure") at an intermediate stage of fabrication is shown according to an exemplary embodiment. FIG. **1** is a cross-sectional view of the structure **100**. The structure **100** may be formed or provided. The structure **100** may include a cell **101** and a cell **103**. The cells **101**, **103**, each includes, for example, an inter-layer dielectric (hereinafter "ILD") **104**, a liner **106**, a lower metal wire **108**, an inter-layer dielectric (hereinafter "ILD") **110**, and a metal cap **112**.

The structure **100** may include several back end of line ("BEOL") layers. In general, the back end of line (BEOL) is the second portion of integrated circuit fabrication where the individual devices (transistors, capacitors, resistors, etc.) are interconnected with wiring on the wafer.

The ILD **104** may be formed by depositing or growing a dielectric material on the BEOL layers, followed by a chemical mechanical polishing (CMP) or etch steps. The ILD **104** may be deposited using typical deposition techniques, for example, atomic layer deposition (ALD), molecular layer deposition (MLD), chemical vapor deposition (CVD), physical vapor deposition (PVD), high density plasma (HDP) deposition, and spin on techniques, followed by a planarization process, such as CMP, or any suitable etch process. In an embodiment, the ILD **104** may include one or more layers. In an embodiment, the ILD **104** may include any dielectric material such as silicon oxide (SiOx), silicon nitride (SiNx), silicon boron carbonitride (SiBCN), NBLoK, a low-k dielectric material (with k<4.0), including but not limited to, silicon oxide, spin-on-glass, a flowable oxide, a high-density plasma oxide, borophosphosilicate glass (BPSG), or any combination thereof or any other suitable dielectric material.

The lower metal wire **108** may be formed by first patterning two or more trenches (not shown) into the ILD **104**, lining the two or more trenches with the liner **106**, and filling the two or more trenches.

The liner **106** separates the conductive interconnect material of the lower metal wire **108** from the ILD **104**. The liner **106** may be composed of, for example, tantalum nitride (TaN), tantalum (Ta), titanium (Ti), titanium nitride (TiN), or a combination thereof. The liner **106** may be deposited utilizing a conventional deposition process such as, for example, CVD, plasma enhanced chemical vapor deposition (PECVD), PVD or ALD. The liner **106** may be 5 nm thick, although a thickness less than or greater than 5 nm may be acceptable. The liner **106** surrounds a lower horizontal surface and a vertical side surface of the lower metal wire **108**.

In an embodiment, the lower metal wire **108** is formed from a conductive material layer which is blanket deposited on top of the structure **100**, and directly on a top surface of the liner **106**, filling the two or more trenches (not shown). The conductive material layer may include materials such as, for example copper (Cu), ruthenium (Ru), cobalt (Co), tungsten (W). The conductive material can be formed by for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), and atomic layer deposition (ALD) or a combination thereof. The lower metal wire **108** is formed by damascene, or patterned from the conductive material layer, using known patterning and etching techniques. There may be any number of openings in the ILD **104**, each filled with the liner **106** and the lower metal wire **108**, on the structure **100**.

A planarization process, such as, for example, chemical mechanical polishing (CMP), may be done to remove excess material from a top surface of the structure **100** such that upper horizontal surfaces of the lower metal wire **108**, the liner **106** and the ILD **104** are coplanar.

In an embodiment, the lower metal wire **108** may have a thickness ranging from about 10 nm to about 100 nm, although a thickness less than 10 nm and greater than 100 nm may be acceptable.

The ILD **110** may be formed as described for the ILD **104**, directly on a top surface of the liner **106**, the lower metal wire **108** and the ILD **104**. The metal cap **112** may be formed by first patterning two or more second trenches (not shown) into the ILD **110** vertically aligned above the lower metal wire **108** and the liner **106**, and filling the two or more second trenches. The ILD **110** is unlikely to contain voids as the ILD **110** is blanket deposited on the planarized upper surface of the lower metal wire **108** and the liner **106**.

In an embodiment, the metal cap **112** is formed from a conductive material layer which is blanket deposited on top of the structure **100**, and directly on a top surface of the ILD **110**, the liner **106**, the lower metal wire **108** and the ILD **104**. The conductive material layer may include materials such as, for example tantalum (Ta), ruthenium (Ru), titanium (Ti), tungsten (W). The conductive material can be formed by for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), and atomic layer deposition (ALD) or a combination thereof. The metal cap **112** is formed by damascene, or patterned from the conductive material layer, using known patterning and etching techniques. Damascene is the method of BEOL interconnect formation. A dielectric is deposited, patterned, and the resulting feature is metallized. Any metal overburden is removed by planarization.

A planarization process, such as, for example, chemical mechanical polishing (CMP), may be done to remove excess material from a top surface of the structure **100** such that upper horizontal surfaces of the metal cap **112** and the ILD **110** are coplanar.

Referring now to FIG. **2**, a cross-sectional view of the structure **100** is shown, according to an embodiment. An inter-layer dielectric (hereinafter "ILD") **116** and a hard mask **118** may be formed.

The ILD **116** may be formed as described for the ILD **104**, directly on upper horizontal surfaces of the metal cap **112** and the ILD **110**. The ILD **116** is unlikely to contain voids as the ILD **116** is blanket deposited on the planarized upper surface of the metal cap **112** and the ILD **110**.

The hard mask **118** may formed on the structure **100** and patterned, directly on an upper horizontal surface of the ILD **116**. The hard mask **118** may be patterned such that portions of the hard mask **118** are removed, forming an opening **120**. The opening **120** may be vertical aligned above the metal cap **112** and the lower metal wire **108**, in each of the cells **101**, **103**.

Referring now to FIG. **3**, a cross-sectional view of the structure **100** is shown, according to an embodiment. Portions of the ILD **116** may be removed. The hard mask **118** may be removed.

The portions of the ILD **116** may be removed exposing an upper surface of the metal cap **112**. The portions of the ILD **116** may be selectively removed using an anisotropic etching technique, such as, for example, reactive ion etching, extending the openings **120**. The portions of the ILD **116** may be etched such that the ILD **116** may have a tapered side surface, with a width of the opening **120** more narrow at a lower surface closer to the metal cap **112** and a width more width at an upper surface of the opening **120**.

The hard mask **118** may be removed using known techniques.

Referring now to FIG. **4**, a cross-sectional view of the structure **100** is shown, according to an embodiment. A bottom electrode **122** may be formed.

In an embodiment, the bottom electrode **122** is formed from a conductive material layer which is blanket deposited on top of the structure **100**, and directly on a top surface of the metal cap **112** and the ILD **116**. The conductive material layer may include materials such as, for example, tantalum nitride (TaN), tantalum (Ta), titanium (Ti), titanium nitride (TiN). The conductive material layer may be deposited using typical deposition techniques, for example, physical vapor deposition, atomic layer deposition, molecular layer deposition, and chemical vapor deposition. A planarization process, such as, for example, chemical mechanical polishing (CMP), may be done to remove excess material from a top surface of the structure **100** such that upper horizontal surfaces of the bottom electrode **122** and the ILD **116** are coplanar.

In an embodiment, the bottom electrode **122** may have a thickness ranging from about 10 nm to about 100 nm, although a thickness less than 10 nm and greater than 100 nm may be acceptable. The bottom electrode **122** may have a tapered side surface, with a width of the bottom electrode **122** at a lower surface closer to the lower metal wire **108**, w2, smaller than a width at an upper surface of the bottom electrode **122**, w1.

Vertical sidewalls of the bottom electrode **122** may have a sidewall angle which may range between 10 and 50 degrees in an embodiment.

Referring now to FIG. **5**, a cross-sectional view of the structure **100** is shown, according to an embodiment. A reference layer **124** may be formed. A hard mask **126** may be formed.

The reference layer **124** may be formed conformally on the structure **100**. The reference layer **124** may cover an upper horizontal surface of the ILD **116** and an upper horizontal surface of the bottom electrode **122**.

The hard mask **126** may formed on the structure **100** and patterned, directly on an upper horizontal surface of the reference layer **124**. The hard mask **126** may be patterned such that portions of the hard mask **126** are removed, and remaining portions of the hard mask **126** are vertically aligned above the bottom electrode **122**, the metal cap **112** and the lower metal wire **108**, in both of the cells **101**, **103**.

Referring now to FIG. **6**, a cross-sectional view of the structure **100** is shown, according to an embodiment. Portions of the reference layer **124** may be removed. The hard mask **126** may be removed.

Portions of the reference layer **124** may be removed selective to the hard mask **126**. The portions of the reference layer **124** may be selectively removed using an anisotropic etching technique, such as, for example, reactive ion etching. The remaining portions of the reference layer **124** may remain vertically aligned above the bottom electrode **122**, the metal cap **112** and the lower metal wire **108**, in both the cells **101**, **103**, and perpendicular to an upper surface of the lower metal wire **108**.

The hard mask **126** may be removed using known techniques.

Referring now to FIG. **7**, a cross-sectional view of the structure **100** is shown, according to an embodiment. A first encapsulation layer **130** may be formed.

The first encapsulation layer **130** may be conformally formed on the structure **100**, on an upper horizontal surface of the ILD **116**, and on an upper horizontal surface and vertical side surfaces of the reference layer **124**. The first encapsulation layer **130** may include materials such as, for example, any dielectric material such as silicon nitride (SiN) and silicon nitride carbon (SiNC) and may include a single layer or may include multiple layers of dielectric material. In an alternate embodiment, the first encapsulation layer **130** may include zirconium oxide (ZrO_2). The first encapsulation layer **130** may be deposited using typical deposition techniques, for example, physical vapor deposition, atomic layer deposition, molecular layer deposition, and chemical vapor deposition. The first encapsulation layer **130** may have a thickness between 3 nm and 30 nm, although thickness greater than 30 nm or less than 3 nm are acceptable.

The first encapsulation layer **130** helps to protect the reference layer from being damaged or oxidized during subsequent ILD materials deposition.

Referring now to FIG. **8**, a cross-sectional view of the structure **100** is shown, according to an embodiment. Portions of the first encapsulation layer **130** may be removed.

The portions of the first encapsulation layer **130** may be selectively removed using an anisotropic etching technique, such as, for example, reactive ion etching. The remaining portions of the first encapsulation layer **130** may remain vertically aligned directly adjacent to the reference layer **124**, in both the cells **101**, **103**. The first encapsulation layer **130** may be removed from upper horizontal surfaces of the reference layer **124** and the ILD **116**.

Referring now to FIG. **9**, a cross-sectional view of the structure **100** is shown, according to an embodiment. An inter-layer dielectric (hereinafter "ILD") **132** may be formed.

The ILD **132** may be formed as described for the ILD **104**, conformally on the structure **100**, covering upper horizontal surfaces of the first encapsulation layer **130**, the reference layer **124** and the ILD **116** and vertical side surfaces of the first encapsulation layer **130**. The ILD **132** is unlikely to contain voids as the ILD **132** is blanket deposited on the structure **100** and fills openings between adjacent reference layers **124** and the first encapsulation layer **130** between cells **101**, **103**. This is less likely to contain voids than forming an inter-layer dielectric surrounding multiple layers of the cell **101**, **103**.

A planarization process, such as, for example, chemical mechanical polishing (CMP), may be done to remove excess material from a top surface of the structure **100** such that upper horizontal surfaces of the ILD **132**, the reference layer **124** and the first encapsulation layer **130** are coplanar.

Referring now to FIG. **10**, a cross-sectional view of the structure **100** is shown, according to an embodiment. A tunneling barrier **136** may be formed. A free layer **138** may be formed. A top electrode **140** may be formed.

The tunneling barrier **136** may be formed conformally on the structure **100**, on upper horizontal surfaces of the ILD **132**, the reference layer **124** and the first encapsulation layer **130**.

The free layer **138** may be formed conformally on the structure **100**, on upper horizontal surfaces of the tunneling barrier **136**.

The top electrode **140** is formed from a conductive material layer which is blanket deposited on top of the structure **100**, and directly on a top surface of the free layer **138**. The conductive material layer may include materials such as, for example, tantalum nitride (TaN), tantalum (Ta), titanium (Ti), titanium nitride (TiN). The conductive material layer may be deposited using typical deposition techniques, for example, physical vapor deposition, atomic layer deposition, molecular layer deposition, and chemical vapor deposition.

Referring now to FIG. **11**, a cross-sectional view of the structure **100** is shown, according to an embodiment. A hard mask **144** may be formed. Portions of the top electrode **140**, portions of the free layer **138** and portions of the tunneling barrier **136** may be removed.

The hard mask **144** may formed on the structure **100** and patterned, directly on an upper horizontal surface of the top electrode **140**. The hard mask **144** may be patterned such that remaining portions of the hard mask **144** are vertically aligned above the reference layer **124**, the bottom electrode **122**, the metal cap **112** and the lower metal wire **108**, in both the cells **101**, **103**.

The portions of the top electrode **140**, the portions of the free layer **138** and the portions of the tunneling barrier **136** are removed selective to the hard mask **144**. Remaining portions of the top electrode **140**, the free layer **138** and the tunneling barrier **136** may have a tapered side surface which is wider at a lower surface, closer to the lower metal wire **108**, and more narrow at an upper surface, further from the lower metal wire **108**.

More specifically, a lower horizontal surface width, w3, of the tunneling barrier **136** is larger than an upper horizontal width, w4, of the tunneling barrier **136**. A lower horizontal surface width, w4, of the free layer **138** is larger than an upper horizontal width, w5, of the free layer **138**. A lower horizontal surface width, w5, of the top electrode **140** is larger than an upper horizontal width, w6, of the top electrode **140**.

Vertical sidewalls of the tunneling barrier **136**, the free layer **138** and the top electrode **140** may each have the same sidewall angle, which may range between 10 and 50 degrees in an embodiment.

Referring now to FIG. **12**, a cross-sectional view of the structure **100** is shown, according to an embodiment. The hard mask **144** may be removed. A second encapsulation layer **146** may be formed.

The second encapsulation layer **146** may be conformably formed on the structure **100**, on an upper horizontal surface of the top electrode **140** and the ILD **132**, and on vertical side surfaces of the top electrode **140**, the free layer **138** and the tunneling barrier **136**.

The second encapsulation layer **146** may include materials such as, for example, any dielectric material such as silicon nitride (SiN) and silicon nitride carbon (SiNC) and may include a single layer or may include multiple layers of dielectric material. The second encapsulation layer **146** may be deposited using typical deposition techniques, for example, physical vapor deposition, atomic layer deposition, molecular layer deposition, and chemical vapor deposition. The second encapsulation layer **130** may have a thickness between 3 nm and 30 nm, although thickness greater than 30 nm or less than 3 nm are acceptable.

The second encapsulation layer **146** helps to protect the free layer from being damaged or oxidized during subsequent ILD materials deposition.

Referring now to FIG. **13**, a cross-sectional view of the structure **100** is shown, according to an embodiment. Portions of the second encapsulation layer **146** may be removed. An inter-layer dielectric (hereinafter "ILD") **148** may be formed.

The portions of the second encapsulation layer **146** may be selectively removed using an anisotropic etching technique, such as, for example, reactive ion etching. The remaining portions of the second encapsulation layer **146** may remain vertically aligned directly adjacent to the top electrode **140**, the free layer **138** and the tunneling barrier **136** in both the cells **101**, **103**. The second encapsulation layer **146** may be removed from upper horizontal surfaces of the top electrode **140** and the ILD **132**.

The ILD **148** may be formed as described for the ILD **104**, conformally on the structure **100**, covering upper horizontal surfaces of the second encapsulation layer **146**, the top electrode **140** and the ILD **132** and vertical side surfaces of the first encapsulation layer **130**.

The ILD **148** is unlikely to contain voids as the ILD **148** is blanket deposited on the structure **100** and fills openings between adjacent top electrodes **140**, free layers **138** and tunneling barriers **136** within the second encapsulation layer **146** between adjacent cells **101**, **103**. This is less likely to contain voids than forming an inter-layer dielectric surrounding multiple layers of the cell **101**, **103**. Additionally, while forming the ILD **148**, the ILD **148** first fills a lower width, w7, between cells **101**, **103**, which is smaller than an upper width, w8, resulting in a lower likelihood of voiding.

Referring now to FIG. **14**, a cross-sectional view of the structure **100** is shown, according to an embodiment. A liner **152** may be formed. An upper metal wire **154** may be formed.

The upper metal wire **154** may be formed by first patterning two or more trenches (not shown) into the ILD **148**, lining the two or more trenches with the liner **152**, and filling the two or more trenches.

The liner **152** separates the conductive interconnect material of the upper metal wire **154** from the ILD **148**. The liner **152** may be formed as described for the liner **106**. The liner **152** surround a lower horizontal surface and a vertical side surface of the upper metal wire **154**.

The upper metal wire **154** may be formed as described for the lower metal wire **108**. There may be any number of openings in the ILD **148**, each lined with the liner **152** and the upper metal wire **154**, on the structure **100**.

A planarization process, such as, for example, chemical mechanical polishing (CMP), may be done to remove excess material from a top surface of the structure **100** such that upper horizontal surfaces of the upper metal wire **154**, the liner **152** and the ILD **148** are coplanar.

The structure **100** has vertically aligned portions of the MTJ stack, including the bottom electrode **122**, the reference layer **124**, the tunneling barrier **136**, the free layer **138** and the top electrode **140**. The MTJ stack is vertically aligned between the lower metal wire **108** and the upper metal wire **154**. The vertical side surfaces of the bottom electrode **122** have been individually formed with a tapered side surface wider at an upper surface. The reference layer **124** has a vertical side surface perpendicular to an upper surface of the lower metal wire **180** and is surrounded by the first encapsulation layer **130**. The tunneling layer **146**, the free layer **138** and the top electrode **140** have been individually formed with a tapered side surface more narrow at an upper surface of each layer, with a consistent angle of the tapered side surface for all three layers. The tunneling layer **146**, the free layer **138** and the top electrode **140** are surrounded by the second encapsulation layer **146**. The resulting side surface of the MTJ stack has an octagon profile.

An MRAM pillar having an octagon profile helps to reduce ILD voiding between adjacent MTJ stacks. This helps to extend scalability of MRAM device memory elements due to void-free gap fill between MRAM pillars and improved embedded MRAM performance due to reduced top contact shorts.

A cross section of each cell **101**, **103** has an octagon profile. The octagon profile of each cell **101**, **103**, includes eight surfaces. The eight surfaces include the lower horizontal surface of the bottom electrode **122**, a first vertical side surface of the bottom electrode **122**, a first vertical side surface of the reference layer **124**, a first combined vertical side surface of the tunneling barrier **136**, the free layer **138** and the top electrode **140**, an upper horizontal surface of the top electrode **140**, a second combined vertical side surface of the top electrode **140**, the free layer **138** and the tunneling barrier **136**, a second vertical side surface of the reference layer **124** and a second vertical side surface of the bottom electrode **122**.

Additionally, the structure **100** has inter-layer dielectric layers formed separately, with the ILD **130** surrounding the bottom electrode **122**, the ILD **132** surrounding the reference layer **124** and the ILD **148** surrounding the first encapsulation layer **146**, the free layer **138** and the top electrode **140**. Each of the ILDs **130**, **132**, **148** are formed individually and have less voiding of an inter-layer dielectric formed at one time for a same total volume of ILD for the entire MTJ stack, for each cell **101**, **103**.

The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiment, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

1. A semiconductor device comprising:

a magnetic tunnel junction (MTJ) stack, wherein a cross section of the MTJ stack comprises an octagon profile;

the MTJ stack comprises vertically aligned layers of a top electrode, a free layer, a tunneling barrier, a reference layer and a bottom electrode;

the bottom electrode comprises a tapered side surface comprising a width at an upper surface of the bottom electrode greater than a width at a lower surface of the bottom electrode;

the reference layer comprises a vertical side surface perpendicular to an upper horizontal surface of the bottom electrode; and

the top electrode, the free layer and the tunneling barrier, each comprise a tapered side surface of the same angle, and each comprise a width at an upper surface less than a width at a lower surface.

2. The semiconductor device according to claim **1**, further comprising:

a first encapsulation layer surrounding vertical side surfaces of the reference layer.

3. The semiconductor device according to claim **1**, further comprising:

a second encapsulation layer surrounding vertical side surfaces of the top electrode, the free layer and the tunneling barrier.

4. The semiconductor device according to claim **1**, further comprising:

a first inter-layer dielectric surrounding the bottom electrode;

a second inter-layer dielectric surrounding the reference layer; and

a third inter-layer dielectric surrounding the top electrode, the free layer and the tunneling barrier.

5. A semiconductor device comprising:

a magnetic tunnel junction (MTJ) stack comprising vertically aligned layers of a top electrode, a free layer, a tunneling barrier, a reference layer and a bottom electrode, wherein a cross section of the vertically aligned layers the MTJ stack comprises an octagon profile, wherein the top electrode, the free layer and the tunneling barrier each comprise a tapered side surface of the same angle, and each comprise a width at an upper surface less than a width at a lower surface; and

a first encapsulation layer surrounding vertical side surfaces of the reference layer.

6. The semiconductor device according to claim **5**, further comprising:

the bottom electrode comprises a tapered side surface comprising a width at an upper surface of the bottom electrode greater than a width at a lower surface of the bottom electrode.

7. The semiconductor device according to claim **5**, further comprising:

the reference layer comprises a vertical side surface perpendicular to an upper horizontal surface of the bottom electrode.

8. The semiconductor device according to claim **5**, further comprising:

a second encapsulation layer surrounding vertical side surfaces of the top electrode, the free layer and the tunneling barrier.

9. The semiconductor device according to claim **5**, further comprising:

a first inter-layer dielectric surrounding the bottom electrode;

a second inter-layer dielectric surrounding the reference layer; and

a third inter-layer dielectric surrounding the top electrode, the free layer and the tunneling barrier.

10. A semiconductor device comprising:

a magnetic tunnel junction (MTJ) stack comprising vertically aligned layers of a top electrode, a free layer, a tunneling barrier, and a reference layer, wherein the top electrode, the free layer and the tunneling barrier each comprise a tapered side surface of the same angle, and each comprise a width at an upper surface less than a width at a lower surface, and wherein the reference layer comprises a vertical side surface perpendicular to an upper horizontal surface of a bottom electrode.

11. The semiconductor device according to claim **10**, further comprising:

a bottom electrode comprising a tapered side surface and a width at an upper surface of the bottom electrode greater than a width at a lower surface of the bottom electrode.

12. The semiconductor device according to claim **10**, further comprising:

a second encapsulation layer surrounding vertical side surfaces of the top electrode, the free layer and the tunneling barrier.

13. The semiconductor device according to claim **10**, further comprising:

a first inter-layer dielectric surrounding a bottom electrode;

a second inter-layer dielectric surrounding a reference layer, and

a third inter-layer dielectric surrounding the top electrode, the free layer and the tunneling barrier.

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